MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR ADVANCED SCHOTTKY TTL,
AND-OR-INVERT GATES, MONOLITHIC SILICON

This amendment forms a part of MIL-M-38510/334B, dated 25 October 1984, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title, delete and substitute as printed above.

1.3, thermal resistance, junction-to-case (θ_{JC}): Delete "0.08°C/W 3/" and substitute "0.08°C/mW 3/".

PAGE 2

1.4: Delete note "2/" and substitute "4/".

Bottom of page: Delete note "2/" and substitute "4/".

3.2.3, delete and substitute:

"3.2.3 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in the specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and will be available upon request."

3.3: Delete "(see 6.5)" and substitute "(see 6.4)".

PAGE 3

TABLE I, I_{IL}, under Unit column delete "µA" and substitute "mA".

PAGE 4

TABLE II, Interim electrical parameters, delete "(pre burn-in)".

* 4.2a(1): Delete "Test condition D or E" and substitute "Test condition D, E, or F."

PAGE 5

4.4.3c(1): After Test condition D, add "E, or F".

4.4.3c(3): Delete "appendix B of MIL-M-38510 and".

The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

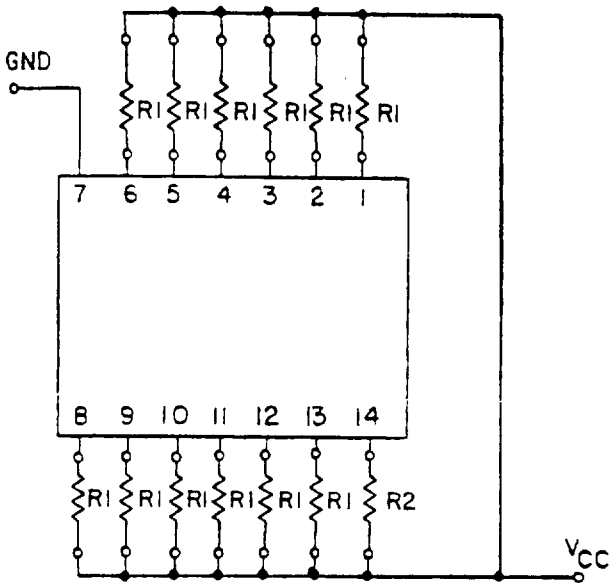
Replacement page	Pages replaced
9	Reprinted without change
10	10

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*Figure 3, add optional burn-in circuit as printed below:

Device 01 excluding case 2
High-temperature (175°C) burn-in (optional circuit).



NOTES:
1. R_1 = 2 KΩ ±5 percent, R_2 = 2Ω
2. V_{CC} shall be high enough to ensure 5.0 V at the device terminal.

FIGURE 3. Burn-in and life test circuit - Continued.

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PAGE 10

*TABLE III, I_{OD}, minimum limits column: Add "60".

*TABLE III, t_{PHL}, test number column: Add "86 - 96" (subgroup 10).

*TABLE III, t_{PLH}, test number column: Add "97 - 107" (subgroup 10).

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from previous amendments were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1039)

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D Case 2 1/ Test no.	1 2 A	2 3 E	3 4 F	4 6 G	5 8 H	6 9 I	7 10 GND	8 12 Y	9 13 J	10 14 K	11 16 B	12 18 C	13 19 D	14 20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = 25°C	V_{OL}	3007	1	2.0 V	GND	GND	GND	GND	GND	GND	20 mA	GND	GND	2.0 V	2.0 V	2.0 V	4.5 V	Y		0.5	V
		"	2	GND	2.0 V	2.0 V	GND	GND	GND	"	"	GND	GND	GND	GND	GND	"	"		"	"
		"	3	GND	GND	GND	2.0 V	2.0 V	2.0 V	"	"	GND	GND	GND	GND	GND	"	"		"	"
		"	4	GND	"	"	GND	GND	GND	"	"	2.0 V	2.0 V	"	"	"	"	"		"	"
	V_{OH}	3006	5	0.8 V	"	"	"	"	"	"	-1 mA	GND	GND	2.0 V	2.0 V	2.0 V	"	"	2.5		"
		"	6	2.0 V	"	"	"	"	"	"	"	"	"	0.8 V	2.0 V	2.0 V	"	"	"		"
		"	7	2.0 V	"	"	"	"	"	"	"	"	"	2.0 V	0.8 V	2.0 V	"	"	"		"
		"	8	2.0 V	"	"	"	"	"	"	"	"	"	2.0 V	2.0 V	0.8 V	"	"	"		"
		"	9	GND	0.8 V	2.0 V	"	"	"	"	"	"	"	GND	GND	GND	"	"	"		"
		"	10	"	2.0 V	0.8 V	"	"	"	"	"	"	"	"	"	"	"	"	"		"
		"	11	"	GND	GND	0.8 V	2.0 V	2.0 V	"	"	"	"	"	"	"	"	"	"		"
		"	12	"	"	"	2.0 V	0.8 V	2.0 V	"	"	"	"	"	"	"	"	"	"		"
		"	13	"	"	"	2.0 V	2.0 V	0.8 V	"	"	"	"	"	"	"	"	"	"		"
		"	14	"	"	"	GND	GND	GND	"	"	0.8 V	2.0 V	"	"	"	"	"	"		"
		"	15	"	"	"	GND	GND	GND	"	"	2.0 V	0.8 V	"	"	"	"	"	"		"
	V_{IC}		16	-18 mA						"							"	A	-1.2	"
			17		-18 mA					"							"	E	"	"
			18			-18 mA				"							"	F	"	"
			19				-18 mA			"							"	G	"	"
			20					-18 mA		"							"	H	"	"
			21						-18 mA	"							"	I	"	"
			22							"		-18 mA					"	J	"	"
			23							"			-18 mA				"	K	"	"
			24							"				-18 mA			"	B	"	"
			25							"					-18 mA		"	C	"	"
			26							"							"	D	"	"
	I_{IH1}	3010	27	2.7 V						"				GND	GND	GND	5.5 V	A	20	μA
		"	28	GND						"				2.7 V	GND	GND	"	B	"	"
		"	29							"				GND	2.7 V	GND	"	C	"	"
		"	30							"				GND	GND	2.7 V	"	D	"	"
		"	31		2.7 V	GND				"							"	E	"	"
		"	32		GND	2.7 V				"							"	F	"	"
		"	33				2.7 V	GND	GND	"							"	G	"	"
		"	34				GND	2.7 V	GND	"							"	H	"	"
		"	35				GND	GND	2.7 V	"							"	I	"	"
		"	36							"		2.7 V	GND				"	J	"	"
		"	37							"		GND	2.7 V				"	K	"	"
	I_{IH2}	"	38	7.0 V						"				GND	GND	GND	"	A	100	"
		"	39	GND						"				7.0 V	GND	GND	"	B	"	"
		"	40	GND						"				GND	7.0 V	GND	"	C	"	"
		"	41	GND						"				GND	GND	7.0 V	"	D	"	"
		"	42		7.0 V	GND				"							"	E	"	"
		"	43		GND	7.0 V				"							"	F	"	"
		"	44				7.0 V	GND	GND	"							"	G	"	"
		"	45				GND	7.0 V	GND	"							"	H	"	"
		"	46				GND	GND	7.0 V	"							"	I	"	"
		"	47							"		7.0 V	GND				"	J	"	"
		"	48							"		GND	7.0 V				"	K	"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.

Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D Case 2 1/ Test no.	1 2 A	2 3 E	3 4 F	4 6 G	5 8 H	6 9 I	7 10 GND	8 12 Y	9 13 J	10 14 K	11 16 B	12 18 C	13 19 D	14 20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = 25°C	I_{IL}	3009	49	0.5 V						GND				5.5 V	5.5 V	5.5 V	5.5 V	A	2/	2/	mA
			50	5.5 V										0.5 V	5.5 V	5.5 V	"	B			
			51	5.5 V										5.5 V	0.5 V	5.5 V	"	C			
			52	5.5 V										5.5 V	5.5 V	0.5 V	"	D			
			53		0.5 V	5.5 V											"	E			
			54		5.5 V	0.5 V											"	F			
			55				0.5 V	5.5 V	5.5 V								"	G			
			56				5.5 V	0.5 V	5.5 V								"	H			
			57				5.5 V	5.5 V	0.5 V								"	I			
			58									0.5 V	5.5 V				"	J			
			59									5.5 V	0.5 V				"	K			
	I_{OS}	3011	60	GND	GND	GND	GND	GND	GND	"	0.0 V	GND	GND	GND	GND	GND	"	Y	-60	-150	"
	I_{OD}		61							"	2.5 V	5.5 V	5.5 V				4.5 V	"			"
	I_{CCH}	3005	62	GND	GND	GND	GND	GND	GND	"		GND	GND	GND	GND	GND	5.5 V	V_{CC}		2.8	"
	I_{CCL}	3005	63	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"		5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	V_{CC}		4.7	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = 125°C and V_C tests are omitted.																				
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_C tests are omitted.																				
9 T_C = 25°C	t_{PHL}	3003 Fig. 4	64	IN	GND	GND	GND	GND	GND	GND	OUT	GND	GND	2.7 V	2.7 V	2.7 V	5.0 V	A to Y	1.5	4.5	ns
			65	2.7 V	"	"	"	"	"	"	"	"	"	IN	2.7 V	2.7 V	"	B to Y	"	"	"
			66	2.7 V	"	"	"	"	"	"	"	"	"	2.7 V	IN	2.7 V	"	C to Y	"	"	"
			67	2.7 V	"	"	"	"	"	"	"	"	"	2.7 V	2.7 V	IN	"	D to Y	"	"	"
			68	GND	IN	2.7 V	"	"	"	"	"	"	"	GND	GND	GND	"	E to Y	"	"	"
			69	"	2.7 V	IN	"	"	"	"	"	"	"	"	"	"	"	F to Y	"	"	"
			70	"	GND	GND	IN	2.7 V	2.7 V	"	"	"	"	"	"	"	"	G to Y	"	"	"
			71	"	"	"	2.7 V	IN	2.7 V	"	"	"	"	"	"	"	"	H to Y	"	"	"
			72	"	"	"	2.7 V	2.7 V	IN	"	"	"	"	"	"	"	"	I to Y	"	"	"
			73	"	"	"	GND	GND	GND	"	"	IN 2.7 V	2.7 V IN	"	"	"	"	J to Y	"	"	"
			74	"	"	"	"	"	"	"	"			"	"	"	"	K to Y	"	"	"
	t_{PLH}		75	IN	"	"	"	"	"	"	"	GND	GND	2.7 V	2.7 V	2.7 V	"	A to Y	"	8.0	"
			76	2.7 V	"	"	"	"	"	"	"	"	"	IN	2.7 V	"	"	B to Y	"	"	"
			77	"	"	"	"	"	"	"	"	"	"	2.7 V	IN	2.7 V	"	C to Y	"	"	"
			78	"	"	"	"	"	"	"	"	"	"	2.7 V	2.7 V	IN	"	D to Y	"	"	"
			79	GND	IN	2.7 V	"	"	"	"	"	"	"	GND	GND	GND	"	E to Y	"	"	"
			80	"	2.7 V	IN	"	"	"	"	"	"	"	"	"	"	"	F to Y	"	"	"
			81	"	GND	GND	IN	2.7 V	2.7 V	"	"	"	"	"	"	"	"	G to Y	"	"	"
			82	"	"	"	2.7 V	IN	2.7 V	"	"	"	"	"	"	"	"	H to Y	"	"	"
			83	"	"	"	2.7 V	2.7 V	IN	"	"	"	"	"	"	"	"	I to Y	"	"	"
			84	"	"	"	GND	GND	GND	"	"	IN 2.7 V	2.7 V IN	"	"	"	"	J to Y	"	"	"
			85	"	"	"	GND	GND	GND	"	"			"	"	"	"	K to Y	"	"	"
10	t_{PHL}	"		Same tests and terminal conditions as for subgroup 9, except T_C= +125°C.															1.0	6.5	"
	t_{PLH}	"																	1.0	8.0	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																				

1/ Pins not referenced (i.e., 1, 5, 7, 11, 15, and 17) are N/C.

2/ I_{IL} limits shall be as follows:

Test	Min/Max limits in mA Circuits A	B	C
I_{IL}	-.25/-.60	-.037-.60	-.037/-.60

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